

Seat No.:

Stony Brook Institute at Anhui University

2024-2025 Fall Semester

***Finite Mathematical Structures* Final Exam**

(Closed-book Exam Duration: 90 minutes)

Section	I	II	III	IV	Total score
Score	15	15	28	42	
Grader					

You may write down the answers of questions 1-10 in the table below. Note that for questions 1-10, you only need to give a short answer and the explanations are not required.

1		6	
2		7	
3		8	
4		9	
5		10	

Section I: Short answer questions (3 points each)

1. What is the probability that the top two cards in a shuffled deck don't form a pair?

Solution:

2. How many ways are there to arrange the seven letters of the word SYSTEMS, when the 3 S's appear consecutively?

Solution:

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3. How many ways are there to distribute 8 different coins into 4 different cups?

Solution:

4. How many ways are there to deal 52 cards to 4 players giving 13 cards to each?

Solution:

5. What is the coefficient of x^{12} in $(x^2 + x^3 + x^4 + \dots)^5$?

Solution:

Section II: Choose the correct option for each question (3 points each)

6. How many different sequences of 15A's and U's are possible in which the third U appears as the twelfth letter in the sequence?

- a) $C(15,2) \cdot 2^3$
- b) $C(11,2) \cdot 2^3$
- c) $P(15,12,3)$
- d) $P(11,2) \cdot 2^3$

7. How many ways are there to form a three-letter sequence using the letters **p, q, r, s, t, u**, without repetition of any letter?

- a) 120
- b) $6!$
- c) 30
- d) $C(6,3)$
- e) $3! C(6,2)$

8. To evaluate $1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^n}$, firstly, we found the generating function $h(x)$, as follows

a) $\frac{3}{1-x}$

b) $\frac{3}{3-x}$

c) $\frac{3}{(1-x)(3-x)}$

d) $\frac{1}{3-x}$

9. How many binary sequences of length 10 are there starting with 0 ?

(Note: A binary sequence is a sequence of zeros and ones (e.g. 010101101))

a) $9!$

b) 9×2^{10}

c) 2^9

d) 10×2^9

10. The number of r -digit sequences with $\{0,1,2,3\}$, where there are

- an even number of 0's
- an odd number of 1's.

a) $a_r = 4^r$

b) $a_r = 4^{r-1}$

c) $a_r = \frac{1}{4!} (4^r)$

d) $a_r = \frac{1}{2} (4^r)$

Section III: Answer and discuss your solution (14 points each)

11. By combinatorial argument, prove the following identity.

$$\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

12. Assume a language center has 100 students with:

- 40 taking French,
- 40 taking Latin,
- 40 taking German,
- 20 taking any given pair of languages,
- 10 taking all 3.

How many students take no language?

Solution:

Section IV: Answer following question (each one 14 points)

13. Find the number of ways to place 21 people into 3 rooms with an even number in each room.

Solution:

14. How many integer solutions are there for the following system

$$x_1 + x_2 + x_3 + x_4 = 20$$

$$0 \leq x_1, x_2, x_3, x_4$$

$$x_1 \leq 10, x_2 \leq 12, x_3 \leq 15, x_4 \leq 18$$

Solution:

15. Build a generating function with $a_r = 3r + 5$

Solution:

Appendix: Some well-known identities

$$(1) \quad \frac{1 - x^{m+1}}{1 - x} = 1 + x + x^2 + \cdots + x^m$$

$$(2) \quad \frac{1}{1 - x} = 1 + x + x^2 + \cdots$$

$$(3) \quad (1 + x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{r}x^r + \cdots + \binom{n}{n}x^n$$

$$(4) \quad (1 - x^m)^n = 1 - \binom{n}{1}x^m + \binom{n}{2}x^{2m} + \cdots + (-1)^k \binom{n}{k}x^{km} + \cdots + (-1)^n \binom{n}{n}x^{nm}$$

$$(5) \quad \frac{1}{(1 - x)^n} = 1 + \binom{1+n-1}{1}x + \binom{2+n-1}{2}x^2 + \cdots + \binom{r+n-1}{r}x^r + \cdots$$

$$(6) \quad \text{If } h(x) = f(x)g(x), \text{ where } f(x) = a_0 + a_1x + a_2x^2 + \cdots \text{ and } g(x) = b_0 + b_1x + b_2x^2 + \cdots, \text{ then}$$

$$h(x) = a_0b_0 + (a_1b_0 + a_0b_1)x + (a_2b_0 + a_1b_1 + a_0b_2)x^2 + \cdots \\ + (a_rb_0 + a_{r-1}b_1 + a_{r-2}b_2 + \cdots + a_0b_r)x^r + \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^r}{r!} + \cdots$$

$$e^{nx} = 1 + nx + \frac{n^2x^2}{2!} + \frac{n^3x^3}{3!} + \cdots + \frac{n^rx^r}{r!} + \cdots$$

$$\frac{1}{2}(e^x + e^{-x}) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots$$

$$\frac{1}{2}(e^x - e^{-x}) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots$$